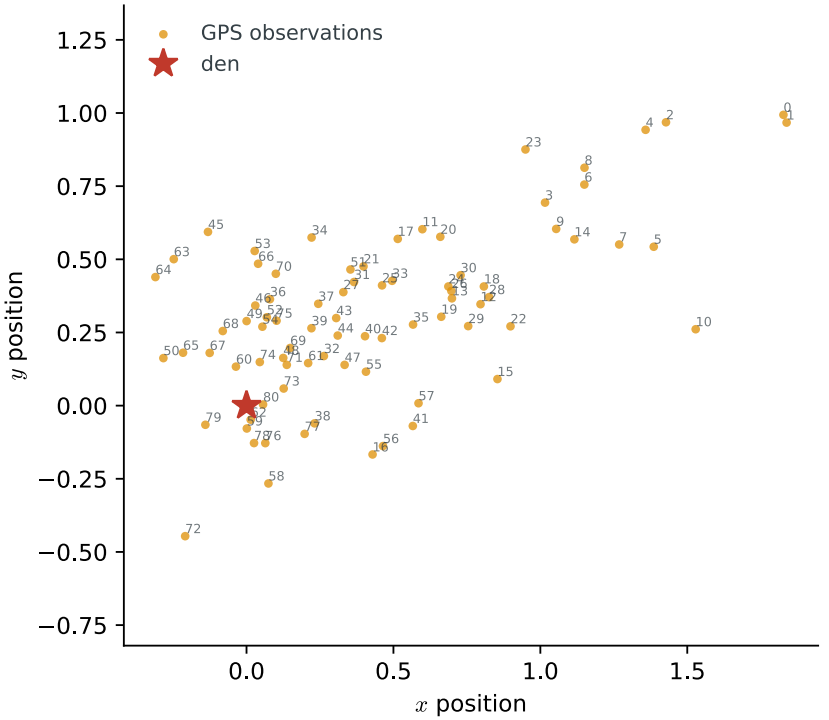
Foundations of Data Science

Lecture 6: Markov chain Monte Carlo

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The normalizing constants of the Kalman filter updates multiply into the likelihood $p(\mathbf{x}_{1:T} \mid \theta)$ of the parameters θ of the wolf model. How do we obtain their posterior $p(\theta \mid \mathbf{x}_{1:T})$?

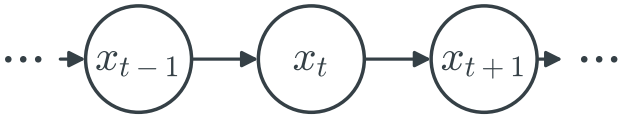


More generally, how to sample from a distribution $p(x)$ when only an unnormalized version $\tilde{p}(x)$ of its density can be evaluated?

Use cases:

- Generating data points x from a generative model $p(x)$.
- Computing expectations $\mathbb{E}_{p(x)}[f(x)]$ or integrals $\int f(x)p(x)dx$.
- Computing posterior distributions $p(\mathbf{z} \mid \mathbf{x})$ in Bayesian inference.

Markov chain Monte Carlo



Markov chains

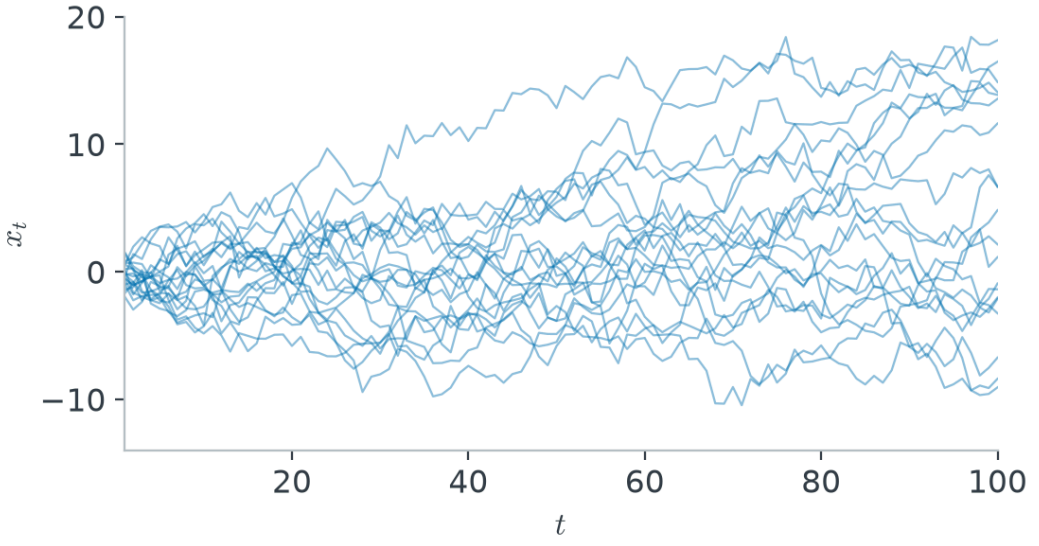
A Markov chain is a sequence of random variables $X_1, X_2, \ldots$ that assumes the conditional independence of X_{t+1} and $X_{1:t-1}$ given X_t.

It follows that the joint distribution of $X_1, \ldots, X_T$ can be factorized as

$$p(x_1, \ldots, x_T) = p(x_1) \prod_{t=1}^{T-1} p(x_{t+1} \mid x_t),$$

where $p(x_1)$ is the **initial distribution** and $p(x_{t+1} \mid x_t)$ is the **transition model**.

When the transition model does not depend on t, i.e., $p(x_{t+1} \mid x_t) = p(x' \mid x)$ for all t, the Markov chain is said to be **time-homogeneous**.



Example 1. Gaussian random walk with $p(x_1) = \mathcal{N}(x_1 \mid 0, 1)$ and transition model $p(x_{t+1} \mid x_t) = \mathcal{N}(x_{t+1} \mid x_t, \sigma^2)$.

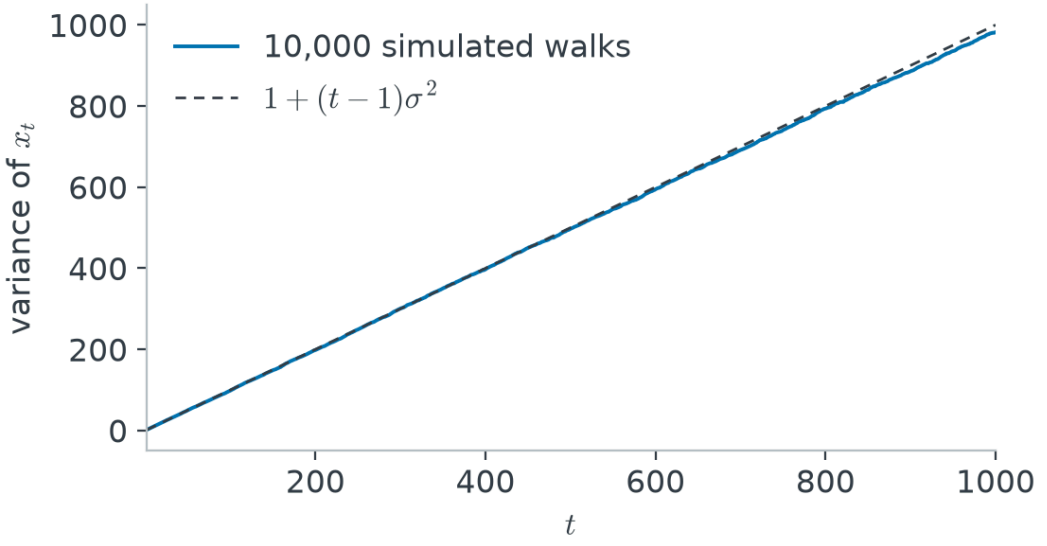
Stationary distributions

A stationary distribution π is a distribution that remains unchanged when passed through the transition model.

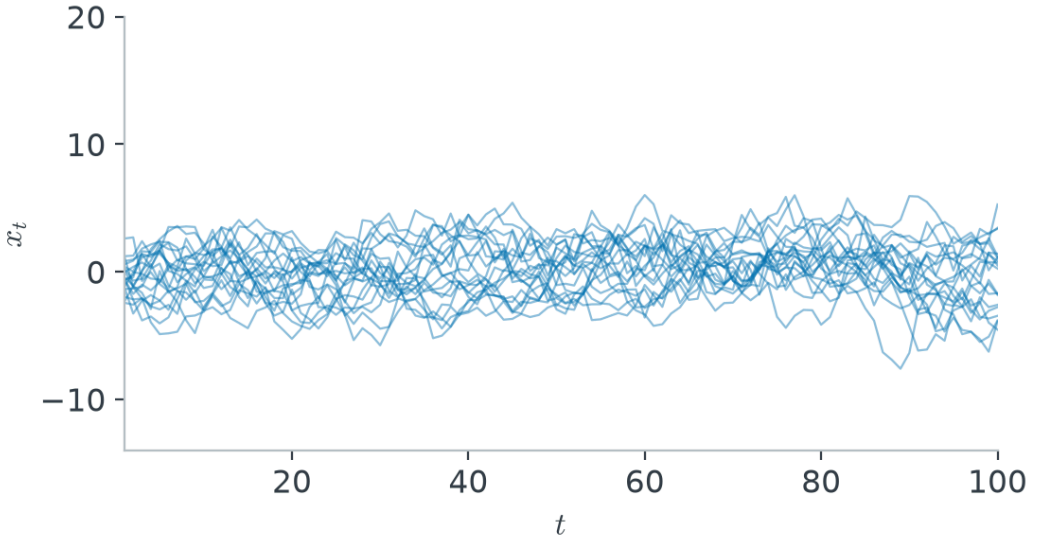
Formally, $\pi(x)$ is stationary for the transition model $p(x' \mid x)$ if

$$\pi(x') = \int \pi(x) p(x' \mid x) dx.$$

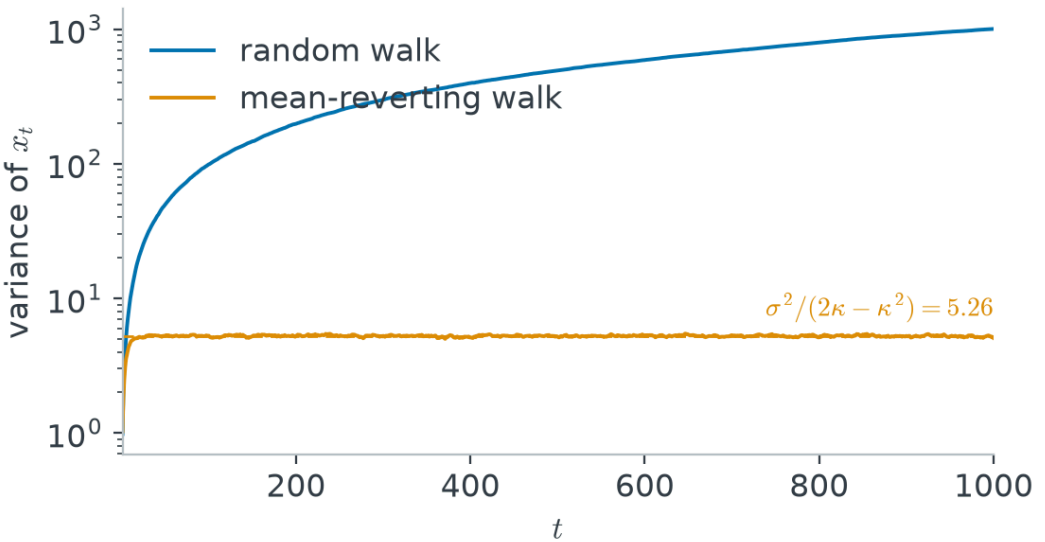
Therefore, if $x_t \sim \pi(x)$, then $x_{t+1} \sim \pi(x)$.



Example 1, continued. The Gaussian random walk does not have a stationary distribution since its variance increases indefinitely over time.



Example 2. Mean-reverting random walk with $p(x_1) = \mathcal{N}(x_1 \mid 0, 1)$ and transition model $p(x_{t+1} \mid x_t) = \mathcal{N}(x_{t+1} \mid x_t - \kappa(x_t - \mu), \sigma^2)$.



For $0 < \kappa < 2$, the mean-reverting random walk has the stationary distribution

$$\pi(x) = \mathcal{N}\left(x \,\middle|\, \mu, \frac{\sigma^2}{2\kappa - \kappa^2}\right).$$

Regularity conditions

Let ν be a distribution on the state space, such as $\mathcal{N}(0, 1)$ on the real line. A set A has positive measure if its probability $\nu(A)$ under ν is positive. A Markov chain is said to be:

- **irreducible** if any set A with positive measure $\nu(A) > 0$ can be reached from any state x, i.e., there exists t such that $p(x_t \in A \mid x_1 = x) > 0$.
- **aperiodic** if it is not forced to cycle through sets, as in $A \to B \to A \to \cdots$.

Basic limit theorem for Markov chains

Theorem. If a time-homogeneous Markov chain has a stationary distribution π and is irreducible and aperiodic, then:

- (Uniqueness) π is its **only** stationary distribution;
- (Convergence) For almost any initial state x_1, $p(x_t) \to \pi(x)$ as $t \to \infty$;
- (Ergodic theorem) For any function f with $\mathbb{E}_\pi[|f(x)|] < \infty$,

$$\lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} f(x_t) = \mathbb{E}_\pi[f(x)]$$

  almost surely.

Proof. Beyond scope in the general case. The next slides prove it under a stronger condition.

Credits: Meyn and Tweedie, Markov Chains and Stochastic Stability, 2009; Tierney, 1994.

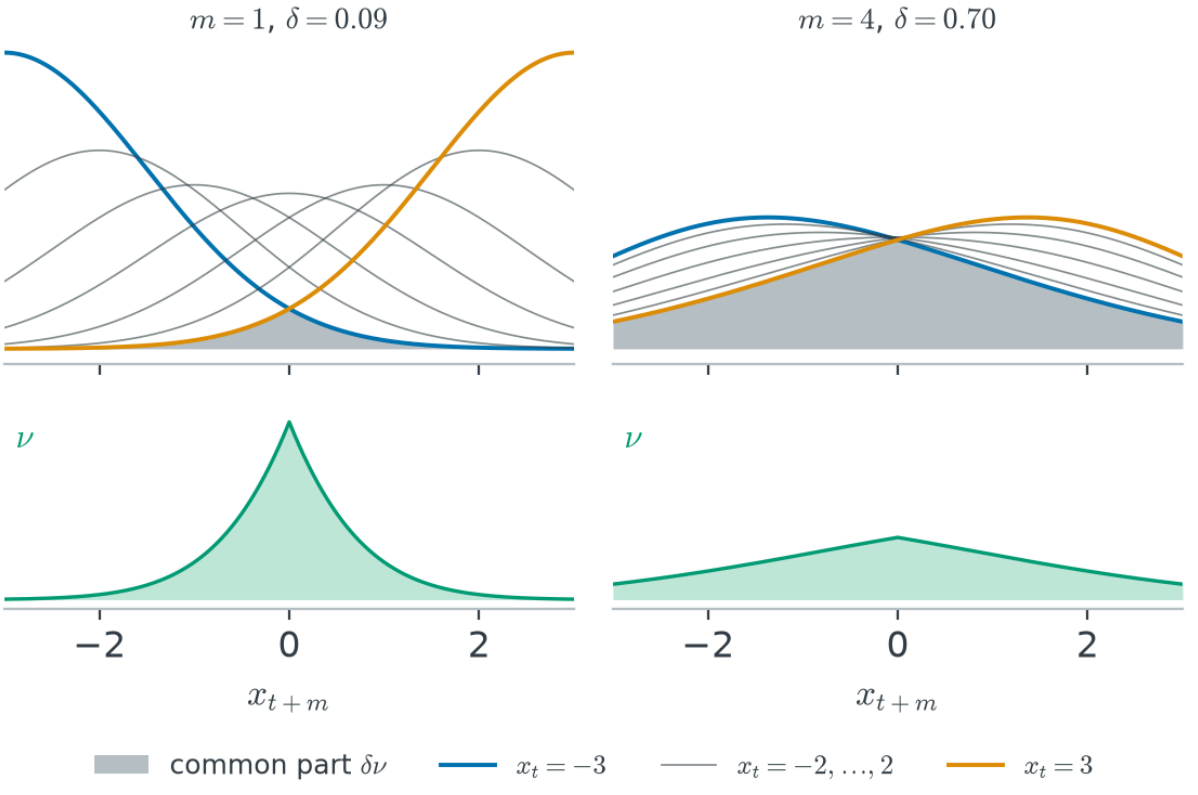
Doeblin condition

A Markov chain satisfies the **Doeblin condition** if there exist $m \geq 1, 0 < \delta < 1$ and a distribution ν such that

$$p(x_{t+m} \in A \mid x_t = x) \geq \delta \nu(A)$$

for all states x and sets A.

Wherever the chain is, the distribution of its state m steps later contains a **common part** $\delta\nu$. A fraction δ of the probability goes to the same places, whatever the current state.



Top: densities of x_{t+m} given $x_t = -3, \ldots, 3$, for a random walk on $[-3, 3]$.
All lie above the shaded common part $\delta\nu$, of area δ, as the condition requires.
Bottom: ν, the common part rescaled to area 1.

Remove the common part from the m-step transition model and rescale what remains:

$$\bar{p}(x_{t+m} \mid x_t) = \frac{p(x_{t+m} \mid x_t) - \delta\nu(x_{t+m})}{1-\delta}.$$

By the Doeblin condition, $\bar{p}$ is non-negative. It integrates to $\frac{1-\delta}{1-\delta} = 1$, since p and ν both integrate to 1. So $\bar{p}$ is a transition model, and

$$p(x_{t+m} \mid x_t) = \delta\nu(x_{t+m}) + (1-\delta)\bar{p}(x_{t+m} \mid x_t).$$

The m-step transition is a mixture. Every m steps, the chain draws its next state from ν with probability δ, whatever its current state, and moves with $\bar{p}$ otherwise.

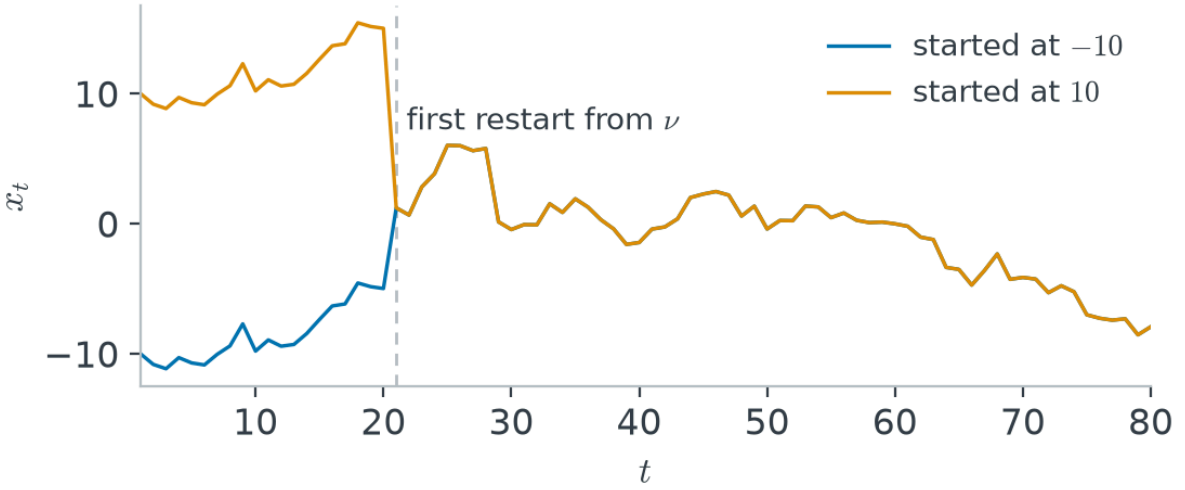
A draw from ν is a **restart**: the new state does not depend on the current one.

Example. The chain

$$p(x' \mid x) = \delta\nu(x') + (1-\delta)\mathcal{N}(x' \mid x, 1),$$

with $\nu = \mathcal{N}(0,1)$ and $\delta = 0.05$, satisfies the Doeblin condition with $m = 1$.

We simulate two copies of this chain from $x_1 = -10$ and $x_1 = 10$, with the same random numbers.



The two copies coincide from the first restart on.

Theorem. Under the Doeblin condition,

- (Existence) the chain has a stationary distribution π;
- (Uniqueness) π is its only stationary distribution;
- (Convergence) for any $p(x_1)$,

$$\|p(x_t) - \pi\|_{\mathrm{TV}} \leq (1-\delta)^{\lfloor (t-1)/m \rfloor},$$

  the probability of no restart in $t-1$ steps, as in the example.

Proof. We first take $m = 1$: the split reads $p(x_{t+1} \mid x_t) = \delta\nu(x_{t+1}) + (1 - \delta)\bar{p}(x_{t+1} \mid x_t)$. Consider two initial distributions p_1 and p_1'. Let $p_1 P$ be the distribution of x_{t+1} when $x_t \sim p_1$, and $p_1 \bar{P}$ likewise when x_{t+1} is drawn from $\bar{p}$. By the split,

$$p_1 P = \delta\nu + (1 - \delta) p_1 \bar{P},$$

so the terms in $\delta\nu$ cancel:

$$\begin{aligned} p_1 P - p_1' P &= \delta\nu + (1 - \delta) p_1 \bar{P} - \delta\nu - (1 - \delta) p_1' \bar{P} \\ &= (1 - \delta)(p_1 \bar{P} - p_1' \bar{P}). \end{aligned}$$

Taking the total variation distance of both sides,

$$\begin{aligned} \|p_1 P - p_1' P\|_{\mathrm{TV}} &= (1 - \delta)\|p_1 \bar{P} - p_1' \bar{P}\|_{\mathrm{TV}} \\ &\leq (1 - \delta)\|p_1 - p_1'\|_{\mathrm{TV}}, \qquad (\star) \end{aligned}$$

since moving two distributions with the same transition model $\bar{p}$ cannot make them more different.

At every step, any two distributions get closer by a factor $1 - \delta$.

A distribution π is stationary if $\pi P = \pi$.

- (Uniqueness) Let π and π' be stationary. By $(\star)$ with $p_1 = \pi$ and $p_1' = \pi'$,

$$\|\pi - \pi'\|_{\mathrm{TV}} = \|\pi P - \pi' P\|_{\mathrm{TV}} \leq (1-\delta)\|\pi - \pi'\|_{\mathrm{TV}}.$$

  A distance at most $1 - \delta < 1$ times itself is zero, so $\pi = \pi'$.

- (Existence) Start from any p_1, and write $p_1 P^n$ for the distribution n steps later. Let $d_n = \|p_1 P^{n+1} - p_1 P^n\|_{\mathrm{TV}}$ be the distance between two consecutive distributions. By $(\star)$ with $p_1 P^n$ and $p_1 P^{n-1}$ in place of p_1 and p_1',

$$\begin{aligned} d_n = \|(p_1 P^n)P - (p_1 P^{n-1})P\|_{\mathrm{TV}} &\leq (1-\delta)\|p_1 P^n - p_1 P^{n-1}\|_{\mathrm{TV}} \\ &= (1-\delta) d_{n-1}. \end{aligned}$$

  Repeating, $d_n \leq (1-\delta)^n d_0 \leq (1-\delta)^n$, since a total variation distance is at most 1. The steps shrink geometrically, so $p_1 P^n$ converges to a limit π. Since $p_1 P^{n+1} = (p_1 P^n)P$ tends both to π and to πP, $\pi P = \pi$.

- (Convergence) The chain starts at x_1, so x_t comes $t-1$ steps later: $p(x_t) = p(x_1)P^{t-1}$. Since π is stationary, $\pi = \pi P^{t-1}$. By $(\star)$ with $p(x_1)P^{n-1}$ and πP^{n-1} in place of p_1 and p_1',

$$\|p(x_1)P^n - \pi P^n\|_{\mathrm{TV}} \leq (1-\delta)\|p(x_1)P^{n-1} - \pi P^{n-1}\|_{\mathrm{TV}}.$$

  Applying this for $n = t-1, t-2, \ldots, 1$,

$$\begin{aligned}
\|p(x_t) - \pi\|_{\mathrm{TV}} &= \|p(x_1)P^{t-1} - \pi P^{t-1}\|_{\mathrm{TV}} \\
&\leq (1-\delta)\|p(x_1)P^{t-2} - \pi P^{t-2}\|_{\mathrm{TV}} \\
&\leq (1-\delta)^2\|p(x_1)P^{t-3} - \pi P^{t-3}\|_{\mathrm{TV}} \leq \ldots \\
&\leq (1-\delta)^{t-1}\|p(x_1) - \pi\|_{\mathrm{TV}} \\
&\leq (1-\delta)^{t-1},
\end{aligned}$$

  since a total variation distance is at most 1.

For $m > 1$, we reduce to the case $m = 1$. Observe the chain every m steps: $y_k = x_{1+km}$. One step of y is m steps of x, so y is a Markov chain with transition P^m. By the Doeblin condition for x, this transition contains the common part $\delta\nu$: y satisfies the condition with $m = 1$.

The previous slides apply to y. Its stationary distributions are those unchanged by m steps, $\pi P^m = \pi$. It remains to carry the results back to x.

- (Uniqueness) A stationary distribution of x is unchanged by m steps, so it is also stationary for y. Since y has only one, so does x.
- (Existence) Let π be the stationary distribution of y, with $\pi P^m = \pi$. To show that $\pi P = \pi$, let $\mu = \pi P$ be π moved one step. Then

$$\mu P^m = (\pi P)P^m = (\pi P^m)P = \pi P = \mu :$$

  μ is also stationary for y, so $\pi = \mu$ by uniqueness for y, and $\pi = \pi P$.

- (Convergence) Write $t-1 = km + r$, with $k = \lfloor (t-1)/m \rfloor$ and $r < m$. Started from $p(x_1)$, the chain y makes k steps in km steps of x, and convergence for y gives

$$\|p(x_1)P^{km} - \pi\|_{\mathrm{TV}} \leq (1-\delta)^k.$$

  The r steps left over move $p(x_1)P^{km}$ and $\pi = \pi P^r$ with the same transition model, which cannot make them more different:

$$\begin{aligned}\|p(x_t) - \pi\|_{\mathrm{TV}} &= \|(p(x_1)P^{km})P^r - \pi P^r\|_{\mathrm{TV}} \\ &\leq \|p(x_1)P^{km} - \pi\|_{\mathrm{TV}} \\ &\leq (1-\delta)^{\lfloor (t-1)/m \rfloor}.\end{aligned}$$

For an irreducible, aperiodic chain with stationary distribution π, $p(x_t)$ approaches π. Once the first states, the **warm-up**, are discarded, the states are (dependent) samples from π.

It remains to build such a chain for a given π (next slides), and to know how long to run it (diagnostics, end of the lecture).



Detailed balance

A transition model $p(x' \mid x)$ satisfies detailed balance with respect to a distribution π if

$$\pi(x)p(x' \mid x) = \pi(x')p(x \mid x')$$

for all x, x'.

Proposition. If a transition model $p(x' \mid x)$ satisfies detailed balance with respect to a distribution π, then π is a stationary distribution of $p(x' \mid x)$.

Proof. Integrating both sides of the detailed balance equation over x, we get

$$\begin{aligned}\int \pi(x)p(x' \mid x)dx &= \int \pi(x')p(x \mid x')dx \\ &= \pi(x') \int p(x \mid x')dx \\ &= \pi(x').\end{aligned}$$

Detailed balance is a **sufficient** condition for stationarity.

It is easier to check than stationarity itself, which makes it the tool of choice to construct a Markov chain with a given stationary distribution.

Metropolis-Hastings

The Metropolis-Hastings algorithm is a procedure for sampling from a target distribution $\pi(x)$, given only an unnormalized version $\tilde{\pi}(x)$ of its density.

The core idea is to construct a Markov chain whose stationary distribution matches the target distribution $\pi(x)$.

The Metropolis-Hastings algorithm proceeds as follows:

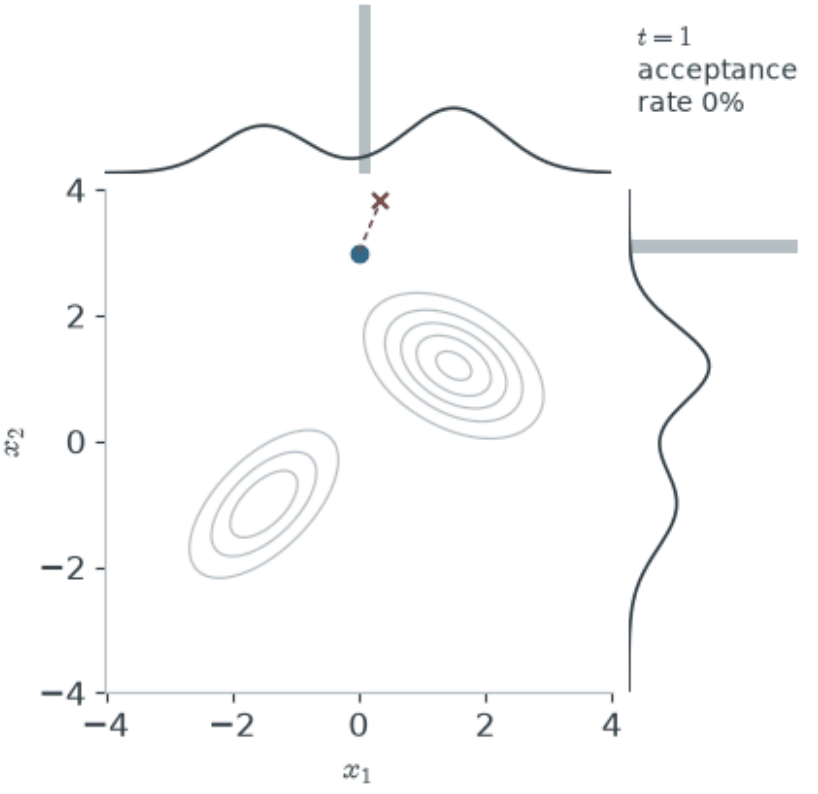
1. Given the current state x_t, propose a new state $x' \sim q(x' \mid x_t)$ using a proposal distribution $q(x' \mid x_t)$.
2. Compute the acceptance ratio

$$\alpha(x' \mid x_t) = \min\left(\frac{\tilde{\pi}(x')q(x_t \mid x')}{\tilde{\pi}(x_t)q(x' \mid x_t)}, 1\right).$$

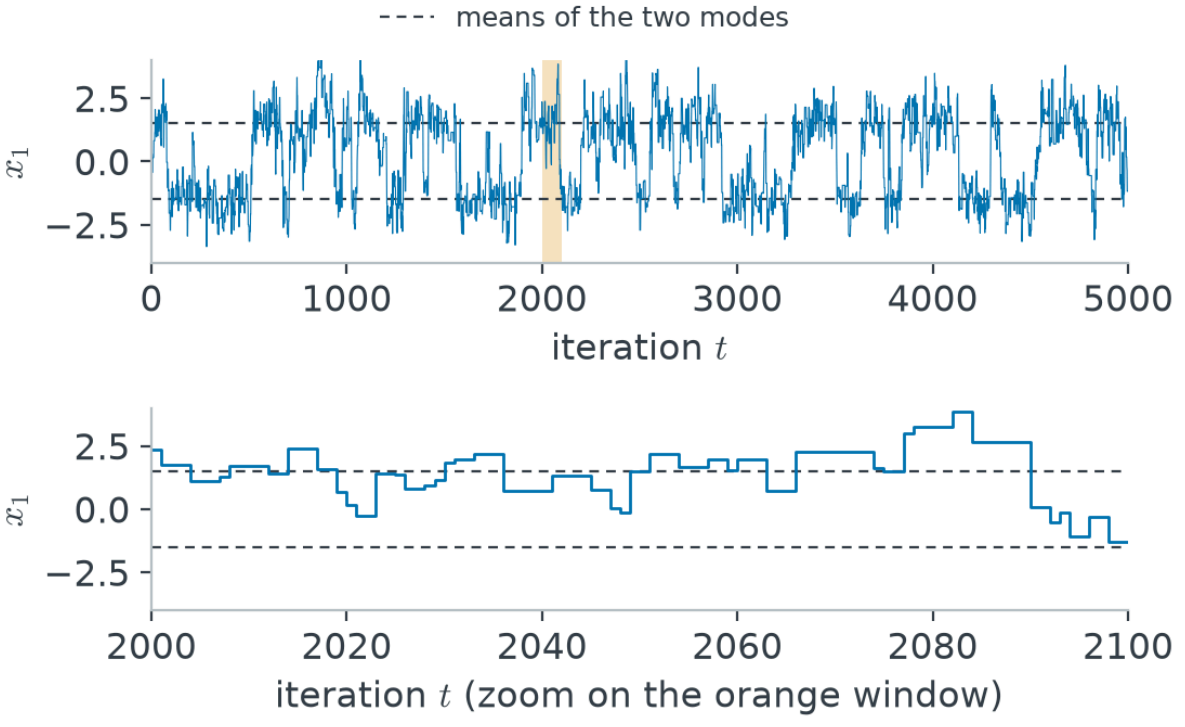
3. Accept the proposed state x' with probability $\alpha(x' \mid x_t)$. If accepted, set $x_{t+1} = x'$. Otherwise, set $x_{t+1} = x_t$.
4. Repeat steps 1-3 for a large number of iterations.

This process generates a sequence of samples $x_1, x_2, \ldots$ that form a Markov chain.

Credits: Metropolis et al., 1953; Hastings, 1970.



Metropolis-Hastings on a mixture of two Gaussians, with the random-walk proposal $q(\mathbf{x}' \mid \mathbf{x}) = \mathcal{N}(\mathbf{x}' \mid \mathbf{x}, \sigma^2 \mathbf{I})$. Proposals are accepted (green) or rejected (red), and the histograms of the chain approach the marginals of the target.



Trace of x_1 along the same chain, and a zoom on 100 iterations.
Each flat segment is a run of rejected proposals.

Proposition. The Metropolis-Hastings algorithm constructs a Markov chain whose stationary distribution is $\pi(x)$.

Proof. The transition model of the Markov chain is given by

$$p(x' \mid x) = \begin{cases} q(x' \mid x)\alpha(x' \mid x) & \text{if } x' \neq x, \\ r(x) & \text{if } x' = x, \end{cases}$$

where $r(x) = 1 - \int q(x' \mid x)\alpha(x' \mid x)dx'$ is the probability of rejecting the proposal and staying at the current state.

For $x' \neq x$, the chain satisfies detailed balance with respect to π:

$$\begin{aligned}
\tilde{\pi}(x)p(x' \mid x) &= \tilde{\pi}(x)q(x' \mid x)\alpha(x' \mid x) \\
&= \tilde{\pi}(x)q(x' \mid x) \min\left(\frac{\tilde{\pi}(x')q(x \mid x')}{\tilde{\pi}(x)q(x' \mid x)}, 1\right) \\
&= \min(\tilde{\pi}(x)q(x' \mid x), \tilde{\pi}(x')q(x \mid x')) \\
&= \tilde{\pi}(x')q(x \mid x') \min\left(\frac{\tilde{\pi}(x)q(x' \mid x)}{\tilde{\pi}(x')q(x \mid x')}, 1\right) \\
&= \tilde{\pi}(x')p(x \mid x').
\end{aligned}$$

Since $\pi(x) = \frac{\tilde{\pi}(x)}{Z}$ for some normalization constant Z, the detailed balance condition also holds for $\pi(x)$. By the proposition on detailed balance, $\pi(x)$ is a stationary distribution of the Markov chain.

For $x' = x$, detailed balance holds trivially.

Proposition. If $q(x' \mid x) > 0$ and $\pi(x) > 0$ for all x and x', as with a Gaussian random-walk proposal, the Metropolis-Hastings chain satisfies the conditions of the basic limit theorem.

Proof.

- Stationarity: π is a stationary distribution, by the previous proposition.
- Irreducibility: for $x' \neq x$, $p(x' \mid x) = q(x' \mid x)\alpha(x' \mid x) > 0$, since both factors are positive. Taking $\nu = \pi$, any set A with positive measure is reached in one step from any state.
- Aperiodicity: since any set with positive measure can be reached in one step from any state, the chain is never forced to cycle through sets.

By the basic limit theorem, the chain converges to π.

Hamiltonian Monte Carlo

The traces show that random-walk proposals explore the target by small steps. Larger steps land in regions of low density and are rejected.

Consider instead a proposal with two properties: it stays on the contour line of $\tilde{\pi}$ through x, so that $\tilde{\pi}(x') = \tilde{\pi}(x)$, and it is symmetric, $q(x \mid x') = q(x' \mid x)$. Both ratios in the acceptance probability are then 1:

$$\alpha(x' \mid x) = \min\left(\frac{\tilde{\pi}(x')}{\tilde{\pi}(x)}\frac{q(x \mid x')}{q(x' \mid x)}, 1\right) = \min(1, 1) = 1.$$

Such moves are always accepted, however far they go. But they never leave their contour line: they cannot visit regions of higher or lower density.

HMC keeps both properties, on pairs: it adds a variable $\mathbf{v}$, and proposes symmetric moves that stay on the contour line of the joint density of $(\mathbf{x}, \mathbf{v})$. Both ratios are again 1, but $\tilde{\pi}(\mathbf{x})$ alone can now go up or down while $\mathbf{v}$ compensates. Redrawing $\mathbf{v}$ jumps to another contour line.

Credits: Duane et al., 1987; Neal, 2011.

Hamiltonian Monte Carlo (HMC) adds an auxiliary variable $\mathbf{v} \in \mathbb{R}^d$, the **momentum**, independent of $\mathbf{x}$, and samples the pairs $(\mathbf{x}, \mathbf{v})$ from

$$\pi(\mathbf{x}, \mathbf{v}) \propto \tilde{\pi}(\mathbf{x}) \mathcal{N}(\mathbf{v} \mid \mathbf{0}, \mathbf{I}) \propto \exp(-H(\mathbf{x}, \mathbf{v})),$$

where the **Hamiltonian** is

$$H(\mathbf{x}, \mathbf{v}) = -\log \tilde{\pi}(\mathbf{x}) + \frac{1}{2} \mathbf{v}^T \mathbf{v}.$$

Since $\mathbf{v}$ is independent of $\mathbf{x}$, discarding $\mathbf{v}$ gives samples of π.

Paths at constant H

On the pairs, a constant density means a constant H. Unlike in $\mathbf{x}$ alone, a move at constant H can change $\tilde{\pi}(\mathbf{x})$: $-\log \tilde{\pi}(\mathbf{x})$ goes up or down, and $\frac{1}{2}\mathbf{v}^T\mathbf{v}$ by the opposite amount.

Along a path $(\mathbf{x}(s), \mathbf{v}(s))$, by the chain rule,

$$\frac{dH}{ds} = \nabla_{\mathbf{x}} H \cdot \frac{d\mathbf{x}}{ds} + \nabla_{\mathbf{v}} H \cdot \frac{d\mathbf{v}}{ds}.$$

Choosing

$$\frac{d\mathbf{x}}{ds} = \nabla_{\mathbf{v}} H = \mathbf{v}, \quad \frac{d\mathbf{v}}{ds} = -\nabla_{\mathbf{x}} H = \nabla \log \tilde{\pi}(\mathbf{x})$$

makes the two terms cancel:

$$\frac{dH}{ds} = \nabla_{\mathbf{x}} H \cdot \nabla_{\mathbf{v}} H - \nabla_{\mathbf{v}} H \cdot \nabla_{\mathbf{x}} H = 0.$$

Along these dynamics, H therefore stays constant.

Each HMC iteration makes two moves on the pair $(\mathbf{x}, \mathbf{v})$. Each move leaves $\pi(\mathbf{x}, \mathbf{v})$ unchanged: if $(\mathbf{x}, \mathbf{v}) \sim \pi(\mathbf{x}, \mathbf{v})$ before it, it still is after.

1. Keep $\mathbf{x}$ and propose a new momentum $\mathbf{v}'$ from $\pi(\mathbf{v}' \mid \mathbf{x}) = \mathcal{N}(\mathbf{v}' \mid \mathbf{0}, \mathbf{I})$, written $\mathcal{N}(\mathbf{v}')$. The change of density is exactly compensated by the proposal, so the acceptance probability is

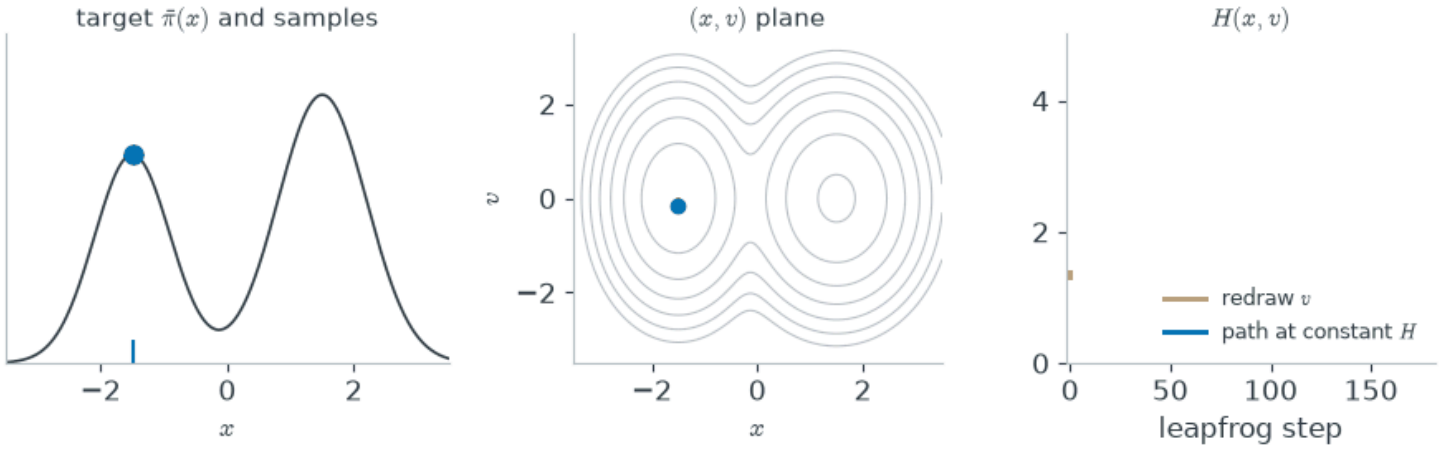
$$\alpha = \min\left(1, \frac{\tilde{\pi}(\mathbf{x})\mathcal{N}(\mathbf{v}')}{\tilde{\pi}(\mathbf{x})\mathcal{N}(\mathbf{v})} \frac{\mathcal{N}(\mathbf{v})}{\mathcal{N}(\mathbf{v}')}\right) = 1.$$

   The candidate is always accepted: $(\mathbf{x}, \mathbf{v}) := (\mathbf{x}, \mathbf{v}')$. H takes a new value.

2. Follow the dynamics from $(\mathbf{x}, \mathbf{v})$, which keep H at this value, and propose the end $(\mathbf{x}', \mathbf{v}')$ of the path. For Metropolis-Hastings over the pairs, with a symmetric proposal, the acceptance probability is

$$\alpha = \min\left(1, \frac{\pi(\mathbf{x}', \mathbf{v}')}{\pi(\mathbf{x}, \mathbf{v})}\right) = \min\left(1, \exp(H(\mathbf{x}, \mathbf{v}) - H(\mathbf{x}', \mathbf{v}'))\right) = 1.$$

   The candidate is always accepted: $(\mathbf{x}, \mathbf{v}) := (\mathbf{x}', \mathbf{v}')$.



Six HMC iterations. Each new v (orange) changes H, each path (blue) keeps it. Only the iteration with the largest H crosses to the other mode.

The dynamics at constant H are simulated with steps of size ϵ. Euler steps let H drift, so long moves would be rejected. The **leapfrog** scheme keeps H nearly constant:

$$\mathbf{v} \leftarrow \mathbf{v} + \frac{\epsilon}{2}\nabla \log \tilde{\pi}(\mathbf{x}), \quad \mathbf{x} \leftarrow \mathbf{x} + \epsilon \mathbf{v}, \quad \mathbf{v} \leftarrow \mathbf{v} + \frac{\epsilon}{2}\nabla \log \tilde{\pi}(\mathbf{x}).$$

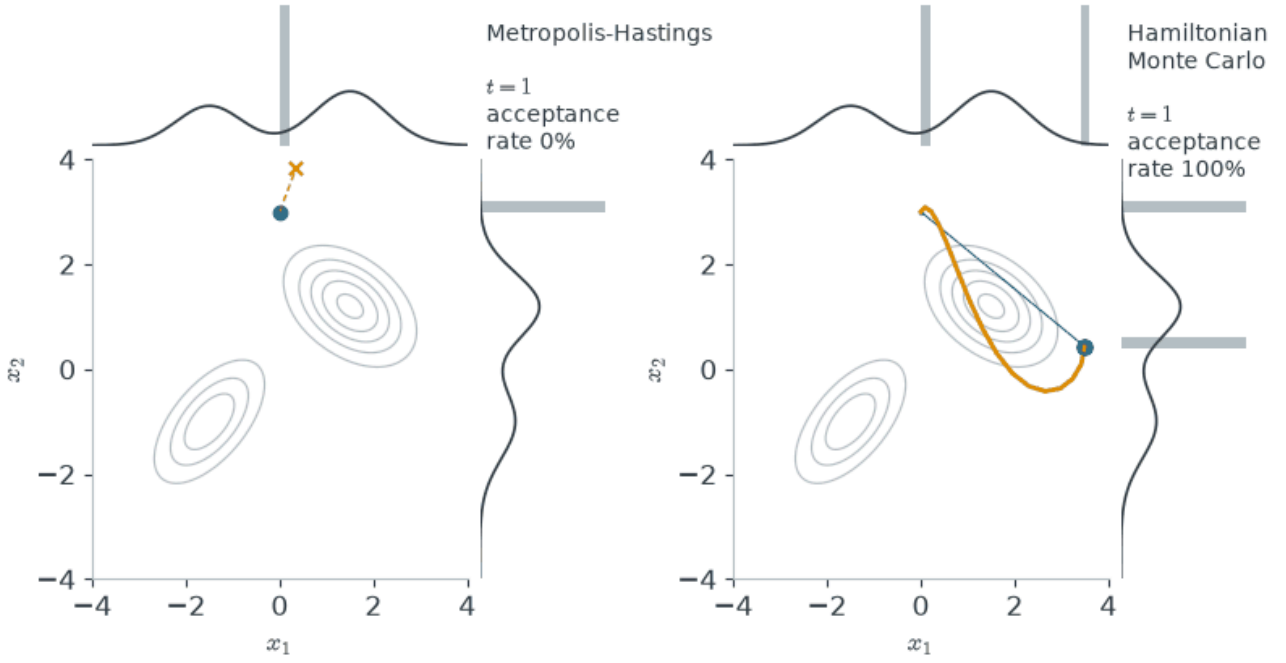
The HMC algorithm proceeds as follows:

1. Given the current state $\mathbf{x}_t$, draw $\mathbf{v} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
2. Starting from $(\mathbf{x}_t, \mathbf{v})$, run L leapfrog steps of size ϵ to obtain $(\mathbf{x}', \mathbf{v}')$.
3. Accept $\mathbf{x}_{t+1} = \mathbf{x}'$ with probability

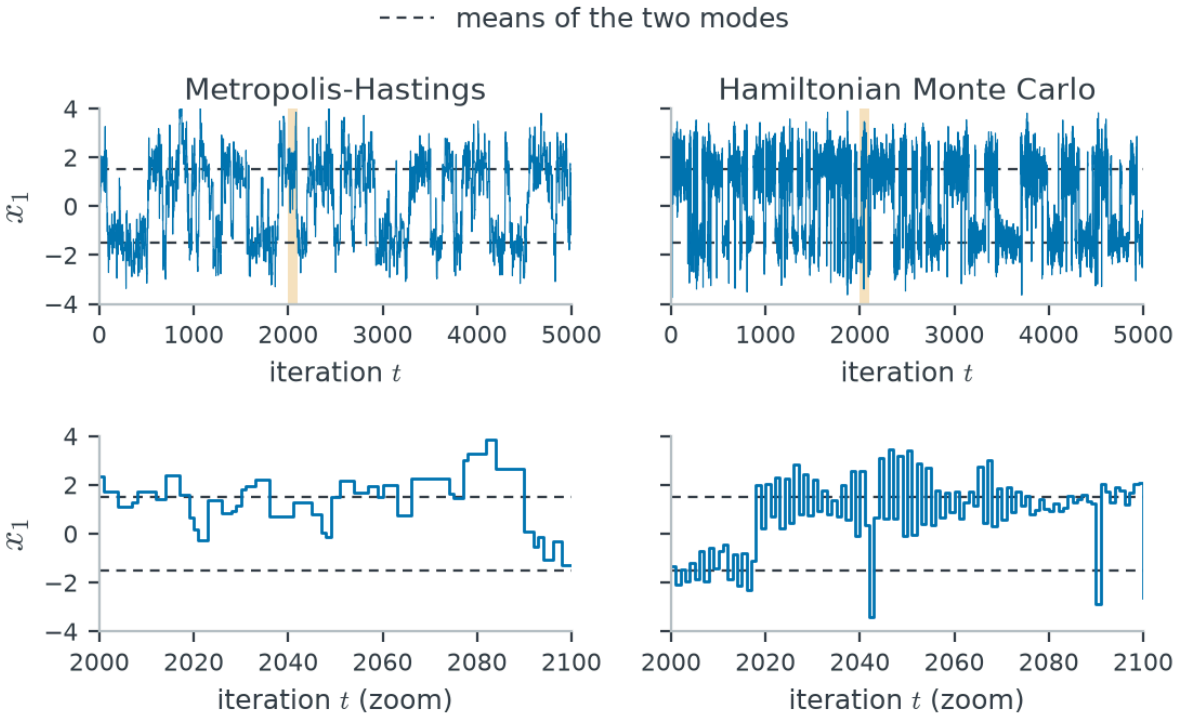
$$\alpha = \min\left(1, \exp(H(\mathbf{x}_t, \mathbf{v}) - H(\mathbf{x}', \mathbf{v}'))\right).$$

   Otherwise, set $\mathbf{x}_{t+1} = \mathbf{x}_t$. In both cases, discard $\mathbf{v}$.

With exact paths, $\alpha = 1$. The leapfrog steps leave a small change of H, which step 3 corrects.



Metropolis-Hastings and HMC on the same target, from the same starting point.
Each HMC iteration follows a leapfrog trajectory (orange).



Traces of x_1 for Metropolis-Hastings and HMC, with zooms on 100 iterations.



Run the first part of `nb06b-mcmc.ipynb` to see Metropolis-Hastings in action.

Bayesian inference with MCMC

Bayesian inference with MCMC

In Bayesian inference, we are often interested in computing the posterior distribution

$$p(\mathbf{z}, \theta \mid \mathbf{x}_{\text{obs}}) = \frac{p(\mathbf{x}_{\text{obs}} \mid \mathbf{z}, \theta) p(\mathbf{z}, \theta)}{p(\mathbf{x}_{\text{obs}})}$$

of latent variables $\mathbf{z}$ and parameters θ given observed data $\mathbf{x}_{\text{obs}}$.

In general, the normalizer

$$p(\mathbf{x}_{\text{obs}}) = \int p(\mathbf{x}_{\text{obs}} \mid \mathbf{z}, \theta) p(\mathbf{z}, \theta) d\mathbf{z} d\theta$$

has no closed form.



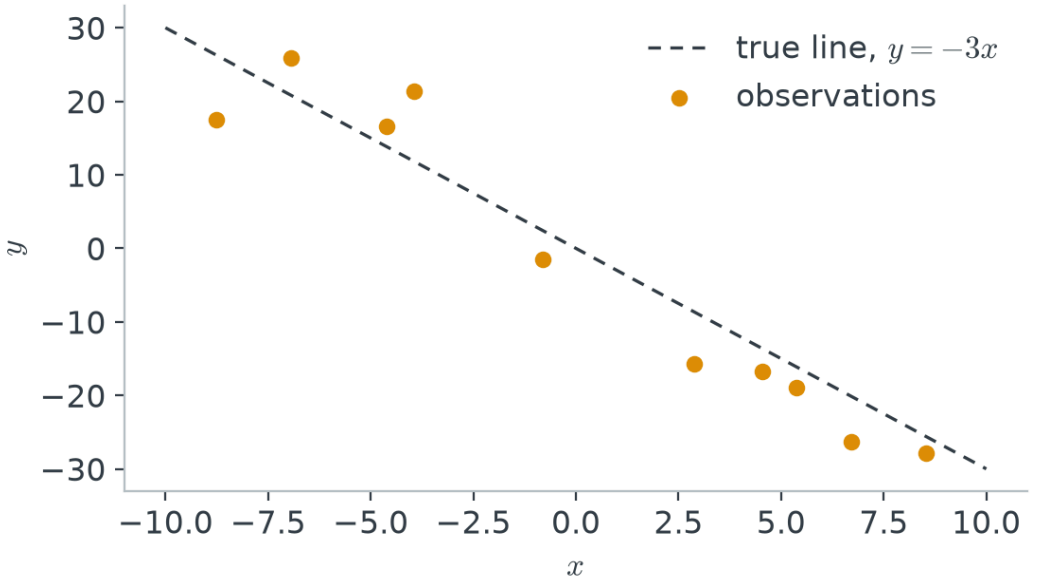
MCMC methods, such as Metropolis-Hastings, generate samples from the posterior $p(\mathbf{z}, \theta \mid \mathbf{x}_{\text{obs}})$ even when the normalizer $p(\mathbf{x}_{\text{obs}})$ has no closed form.

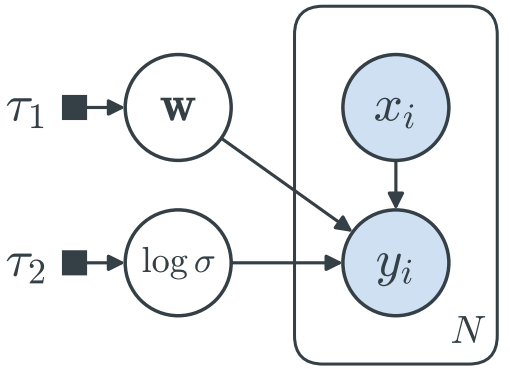
MCMC methods only require an unnormalized density. The **joint density**

$$p(\mathbf{x}_{\text{obs}}, \mathbf{z}, \theta) = p(\mathbf{x}_{\text{obs}} \mid \mathbf{z}, \theta) p(\mathbf{z}, \theta)$$

can therefore be used as the unnormalized target $\tilde{\pi}$ of Metropolis-Hastings, without the normalizer $p(\mathbf{x}_{\text{obs}})$.

Example 3. Bayesian linear regression. Consider a data set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ of N pairs of an input $x_i \in \mathbb{R}$ and a target $y_i \in \mathbb{R}$.





We model y given x as a line with Gaussian noise,

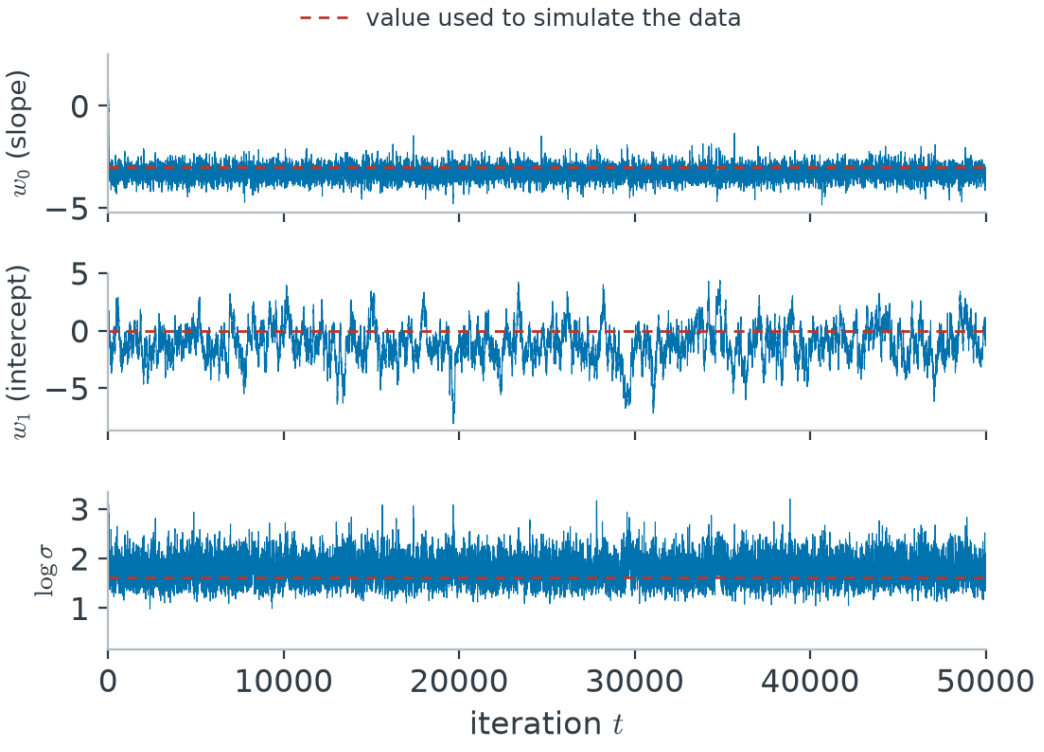
$$p(y \mid x, \mathbf{w}, \log \sigma) = \mathcal{N}(y \mid w_0 x + w_1, \sigma^2),$$

where $\mathbf{w} = (w_0, w_1)$ holds the slope and the intercept. The priors are $p(\mathbf{w}) = \mathcal{N}(\mathbf{w} \mid \mathbf{0}, \tau_1^2 \mathbf{I})$ and $p(\log \sigma) = \mathcal{N}(\log \sigma \mid 0, \tau_2^2)$, with hyperparameters τ_1 and τ_2.

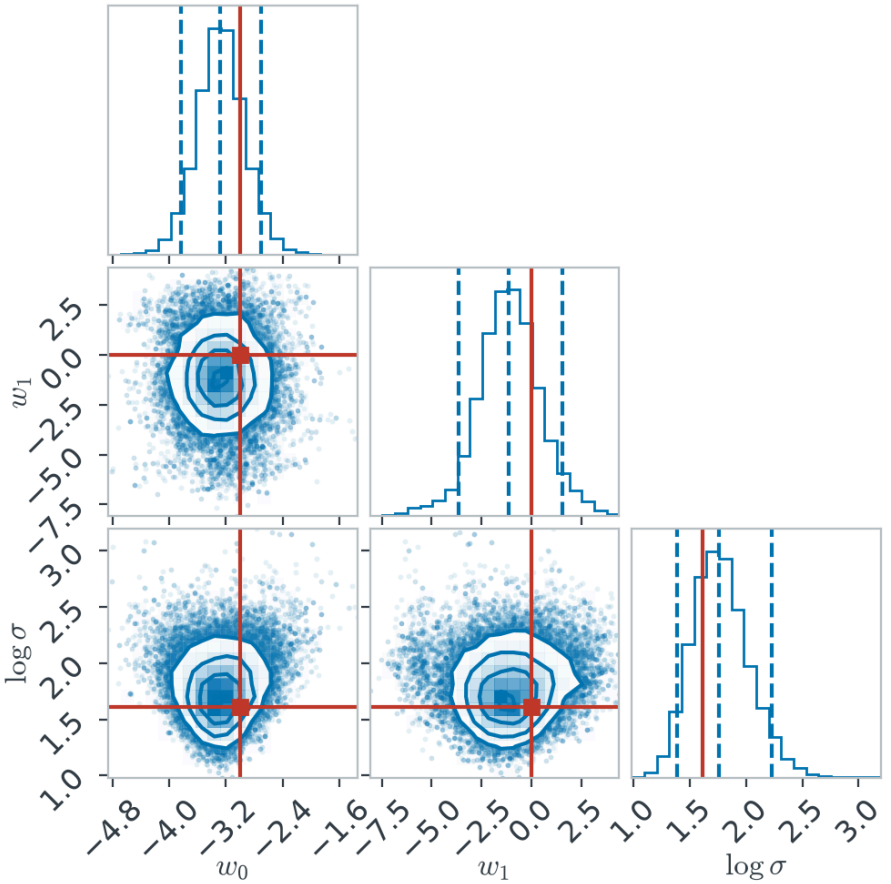
The target distribution is the posterior $p(\mathbf{w}, \log \sigma \mid \mathcal{D})$, which is proportional to the joint density

$$p(\mathbf{w}, \log \sigma \mid \mathcal{D}) \propto \left(\prod_{i=1}^{N} p(y_i \mid x_i, \mathbf{w}, \log \sigma) \right) p(\mathbf{w}) p(\log \sigma).$$

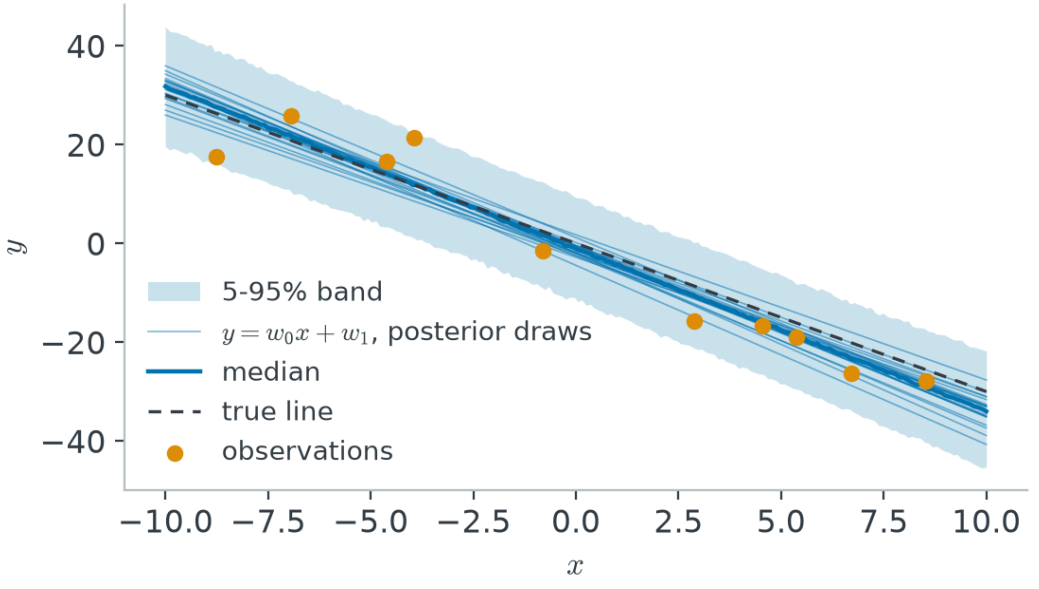
We can use Metropolis-Hastings to sample from this posterior distribution using the joint density as the unnormalized target density.



Traces of w_0, w_1 and $\log \sigma$ along the Markov chain of Metropolis-Hastings.



Posterior samples, after a warm-up of 1000 iterations.



Posterior predictive distribution $p(y \mid x, \mathcal{D})$, median and 5-95% band, with 20 lines $y = w_0 x + w_1$ drawn from the posterior.



Run the end of `nb06b-mcmc.ipynb` to sample the posterior of the wolf's parameters.

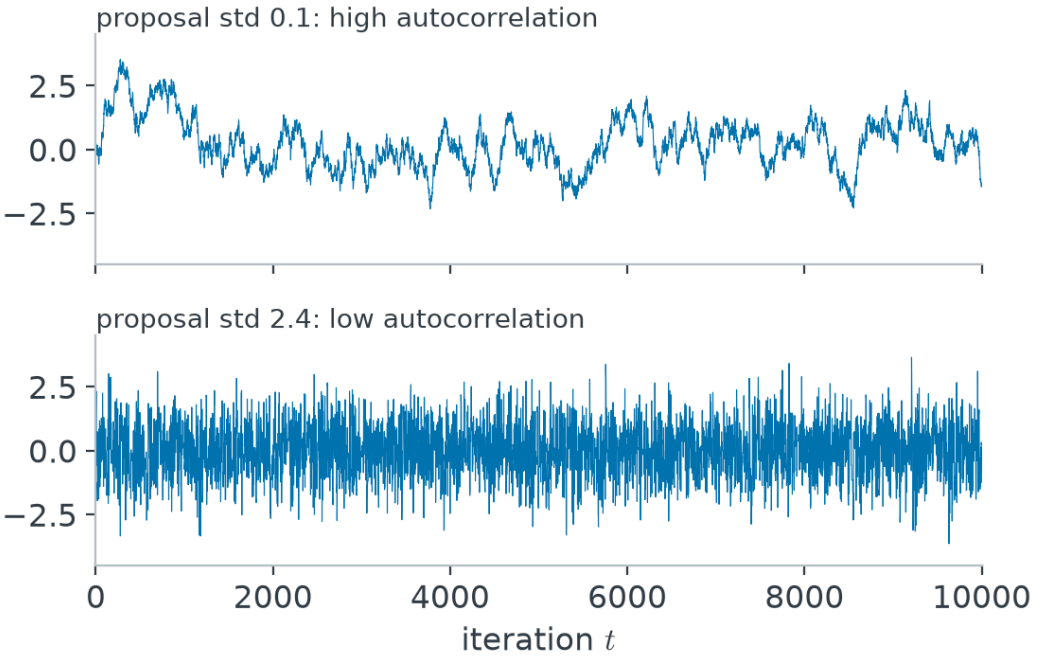


Diagnostics for MCMC

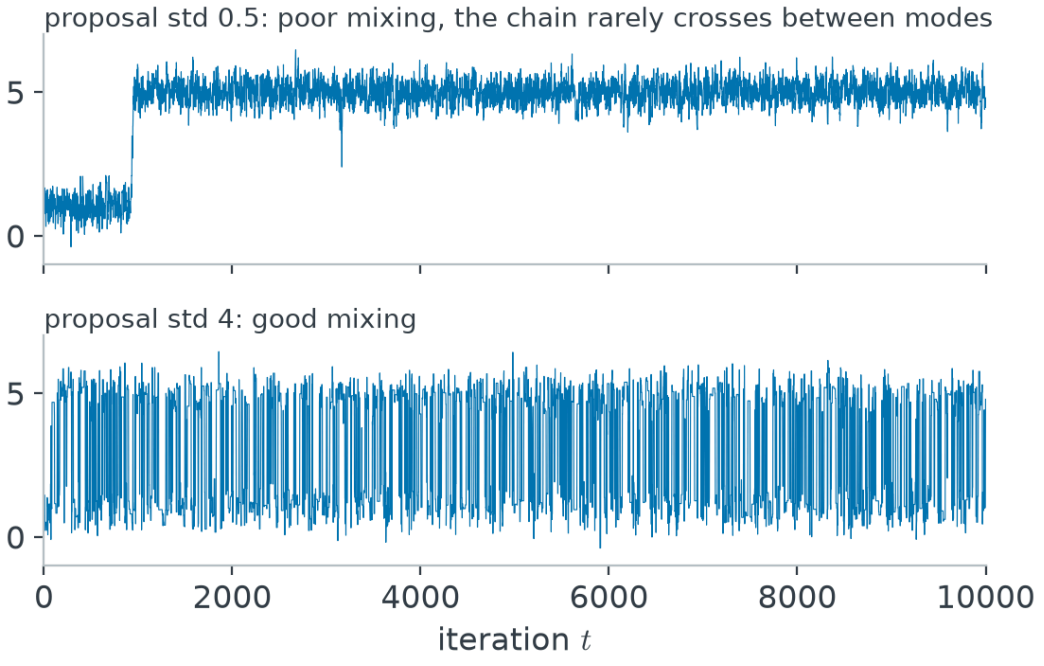
The limit theorem guarantees that a chain converges to the target, but not how fast. Diagnostics check the chains themselves in terms of **autocorrelation**, **mixing** and **convergence**.

Trace plots of the Markov chains can reveal mixing, autocorrelation, and convergence issues:

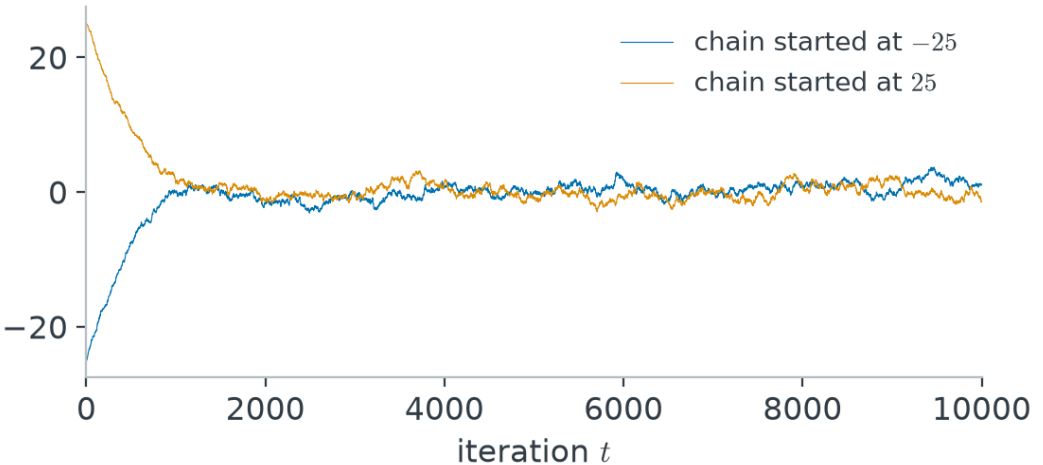
- High autocorrelation is indicated by chains that exhibit strong correlations between successive samples.
- Poor mixing is indicated by chains that explore the state space slowly or get stuck in certain regions.
- When running multiple chains, lack of convergence is indicated by chains that do not overlap or converge to the same distribution.



High vs. low autocorrelation. Highly autocorrelated chains mix poorly.



Poor vs. good mixing for a multimodal target. Poor mixing can be due to small proposal steps or energy barriers between modes.



Two chains run on the same target but initialized differently.
Traces show convergence to the same distribution.

Acceptance rate

The first quantitative diagnostic to monitor is the **acceptance rate**, defined as the fraction of proposed samples that are accepted.

- A very low acceptance rate (e.g., < 10%) indicates that the proposal distribution is making too large jumps, leading to many rejections.
- A very high acceptance rate (e.g., > 80%) suggests that the proposal distribution is making too small jumps, resulting in slow exploration of the state space.

Effective sample size

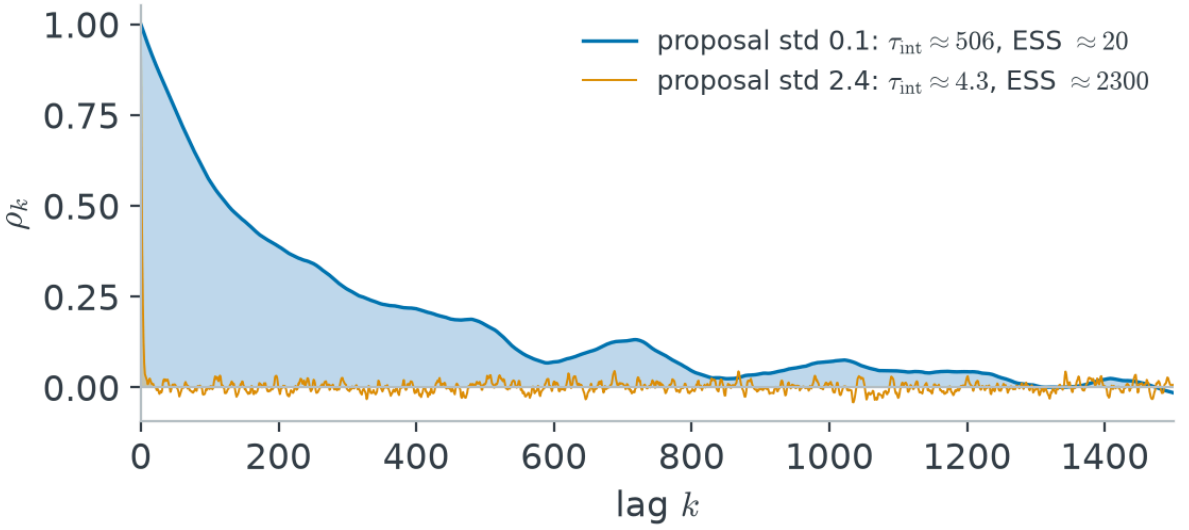
By the ergodic theorem, the average of $f(x_t)$ along the chain converges to $\mathbb{E}_\pi[f(x)]$. Its precision depends on the autocorrelation at lag k,

$$\rho_k = \frac{\mathrm{Cov}(f(x_t), f(x_{t+k}))}{\mathrm{Var}(f(x_t))}.$$

The integrated autocorrelation time $\tau_{\mathrm{int}} = 1 + 2\sum_{k=1}^{\infty} \rho_k$ sets the variance of the average of T samples:

$$\mathrm{Var}\left[\frac{1}{T}\sum_{t=1}^{T} f(x_t)\right] \approx \frac{\tau_{\mathrm{int}}}{T}\mathrm{Var}(f(x_t)).$$

The **effective sample size** $\mathrm{ESS} = T/\tau_{\mathrm{int}}$ is the number of independent samples whose average has the same variance.



Autocorrelation of two random-walk chains of 10000 iterations.

τ_{int} is one plus twice the shaded area.

Gelman-Rubin statistic

Run M independent chains of T iterations from dispersed starting points. Once they have all converged to π, the chains look alike, and the variance within a chain matches the variance of all chains pooled.

The **Gelman-Rubin statistic** compares the two:

$$\hat{R} = \sqrt{\frac{\frac{T-1}{T}W + \frac{1}{T}B}{W}},$$

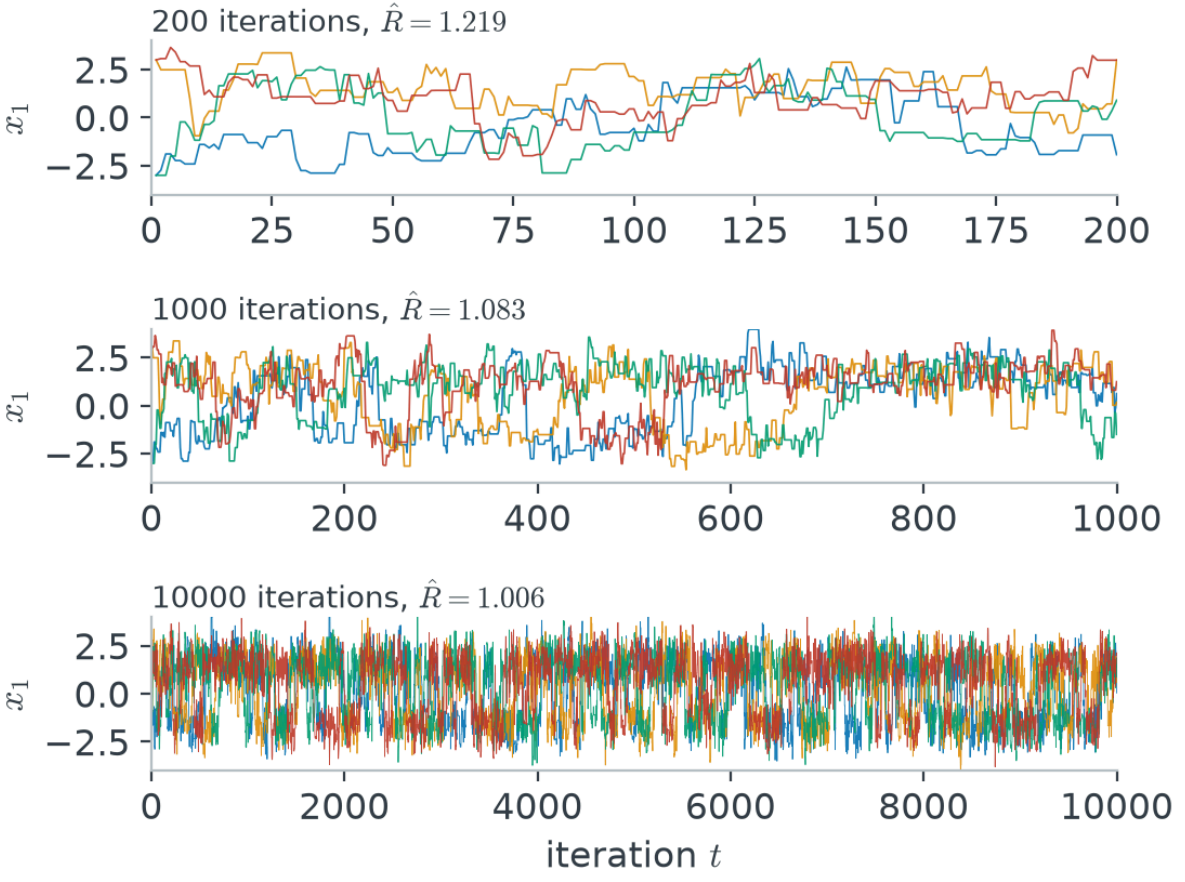
where the **within-chain variance** W and the **between-chain variance** B are

$$W = \frac{1}{M}\sum_{j=1}^{M} s_j^2, \quad B = \frac{T}{M-1}\sum_{j=1}^{M}(\bar{x}_j - \bar{x})^2,$$

with s_j^2 and $\bar{x}_j$ the variance and the mean of chain j, and $\bar{x}$ the mean of all draws.

$\hat{R}$ approaches 1 as the chains agree.

Credits: Gelman and Rubin, 1992.



Four Metropolis-Hastings chains on the two-mode target, started at its corners.

In practice, chains are run until $\hat{R} < 1.01$.

Credits: Vehtari et al., 2021.

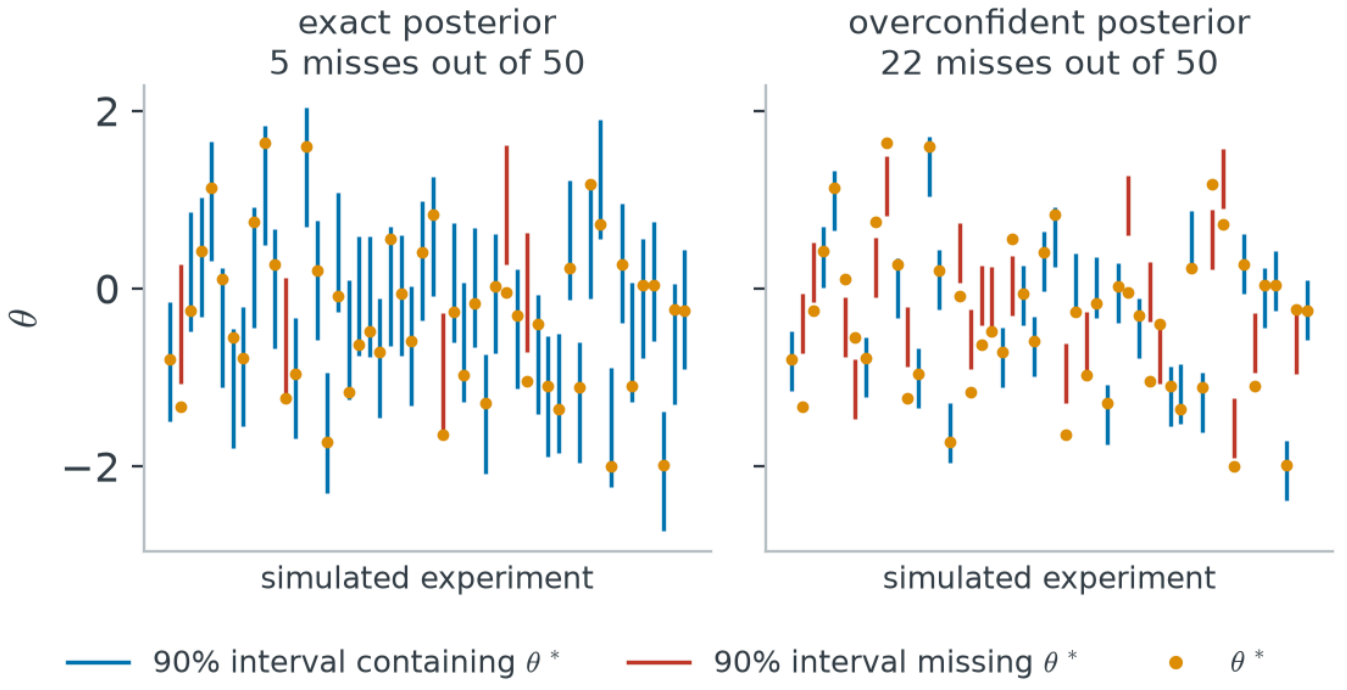
Calibration

Diagnostics cannot detect an error in the log density, the sampler or the posterior approximation.

In simulated experiments, draw $\theta^* \sim p(\theta)$, simulate $\mathcal{D} \sim p(\mathcal{D} \mid \theta^*)$, and compute a credible region $C_\alpha(\mathcal{D})$ of level α. A posterior is **calibrated** if

$$P(\theta^* \in C_\alpha(\mathcal{D})) = \alpha.$$

The exact posterior is calibrated, so a sampler that is not calibrated is wrong. Calibration is necessary, but not sufficient: the prior is calibrated too.



90% credible intervals of a Gaussian mean, in 50 simulated experiments.
The exact posterior misses θ^* in about 10% of them.

Simulation-based calibration

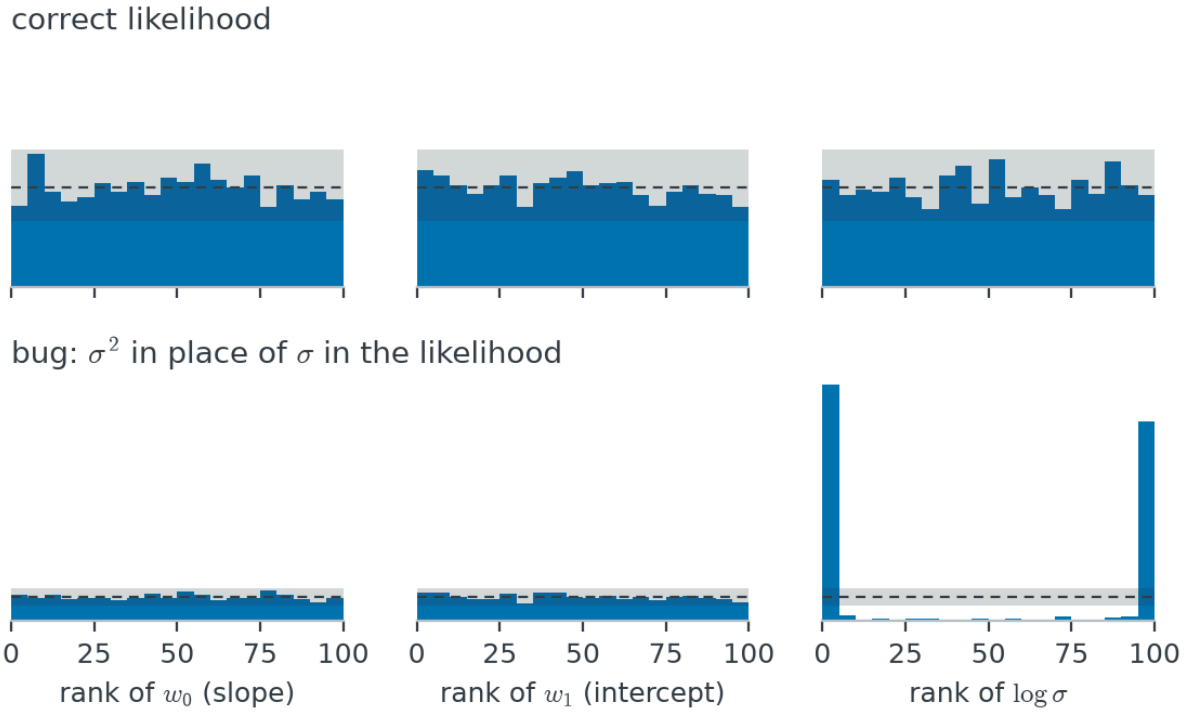
Simulation-based calibration tests whether a sampler is calibrated. For a scalar parameter θ, repeat for $s = 1, \ldots, S$:

1. Draw $\theta_s^* \sim p(\theta)$ and $\mathcal{D}_s \sim p(\mathcal{D} \mid \theta_s^*)$.
2. Draw L samples $\theta_1, \ldots, \theta_L$ with the sampler targeting $p(\theta \mid \mathcal{D}_s)$.
3. Compute the rank $r_s = \sum_{l=1}^{L} 1[\theta_l < \theta_s^*] \in \{0, \ldots, L\}$, where $1[\cdot]$ is 1 if the condition holds and 0 otherwise.

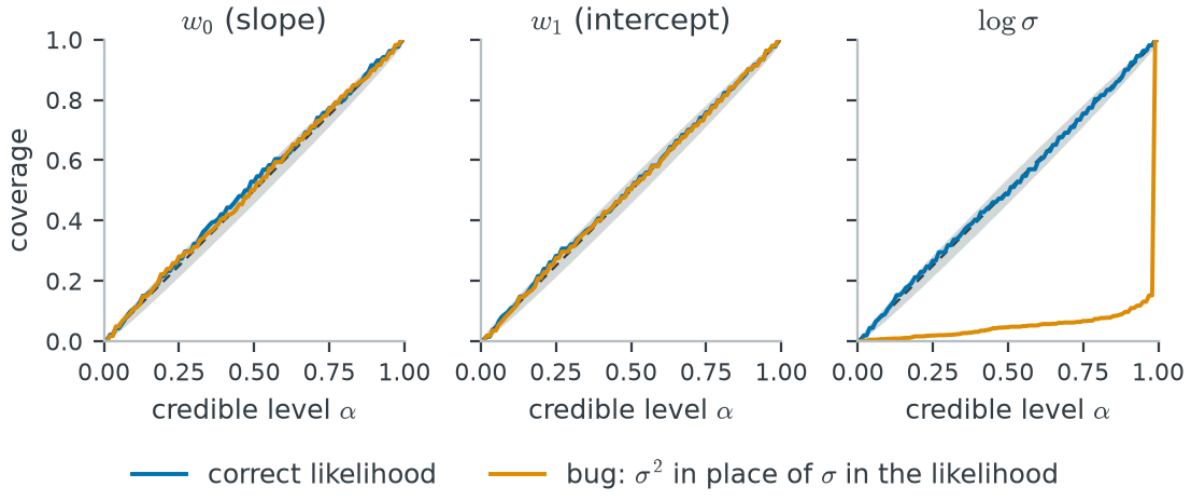
Proposition. If the sampler draws independent samples from the exact posterior, the ranks r_s are uniformly distributed over $\{0, \ldots, L\}$.

The true value θ_s^* lies in the central credible interval of level α when its rank lies in the central fraction α of $\{0, \ldots, L\}$. Uniform ranks make this happen with probability α, for every α: the sampler is calibrated.

Credits: Talts et al., 2018.



Rank histograms of simulation-based calibration for the Bayesian linear regression. The grey band holds 99% of the counts of uniform ranks.



Coverage of the central credible intervals of level α, read from the same ranks.

A calibrated sampler follows the diagonal, $P(\theta^* \in C_\alpha(\mathcal{D})) = \alpha$.

The end.